

Desigualdad maximal de Kolmogorov

Teorema 1 (desigualdad maximal de Kolmogorov). Sean $\{X_j\}_{j \in \mathbb{N}_n}$ variables aleatorias independientes y centradas (i.e. $\forall j \in \mathbb{N} : \mathbb{E}[X_j] = 0$) y $\forall j \in \mathbb{N}_n : X_j \in \mathcal{L}^2(\mathbb{P})$

$$\implies \forall t \in \mathbb{R} : \mathbb{P} \left(\max_{1 \leq k \leq n} |S_k| > t \right) \leq \frac{\mathbb{V}(S_n)}{t^2} \quad \text{donde} \quad S_k = \sum_{j=1}^k X_j.$$

Demostración: Definimos $\forall k \in \mathbb{N}_n, t \in \mathbb{R}_+$:

$$A_k(t) = \{\omega \in \Omega : |S_k(\omega)| > t \wedge \forall j < k : |S_j(\omega)| \leq t\}.$$

Entonces, $\left\{ \omega \in \Omega : \max_{1 \leq k \leq n} |S_k(\omega)| > t \right\} = \bigsqcup_{k=1}^n A_k(t)$ y, por tanto,

$$\begin{aligned} \mathbb{V}(S_n) &= \mathbb{E}[S_n^2] = \int_{\Omega} S_n^2(\omega) d\mathbb{P}(\omega) \geq \int_{\{\max_{1 \leq k \leq n} |S_n| > t\}} S_n^2(\omega) d\mathbb{P}(\omega) = \sum_{k=1}^n \int_{A_k(t)} S_n^2(\omega) d\mathbb{P}(\omega) = \\ &\geq \sum_{k=1}^n \int_{A_k(t)} (S_n - S_k + S_k)^2 d\mathbb{P} = \sum_{k=1}^n \int_{A_k(t)} S_k^2 + 2S_k(S_n - S_k) + (S_n - S_k)^2 d\mathbb{P} \\ &\geq \sum_{k=1}^n \int_{A_k(t)} S_k^2 d\mathbb{P} + \sum_{k=1}^n 2 \int_{\Omega} \underbrace{\mathbb{1}_{A_k(t)} \cdot S_k}_{\xrightarrow{0} (*)} (S_n - S_k) d\mathbb{P} \\ &\geq \sum_{k=1}^n t^2 \cdot \mathbb{P}(A_k(t)) = t^2 \cdot \mathbb{P} \left(\max_{1 \leq k \leq n} S_k \right) \end{aligned}$$

(*) Tenemos que $\mathbb{1}_{A_k(t)} \cdot S_k$ es $\sigma(X_1, \dots, X_k)$ -medible y, además, $S_n - S_k = \sum_{j=k+1}^n X_j$ es $\sigma(X_{k+1}, \dots, X_n)$ -medible. Por el teorema Π - λ , $\mathbb{1}_{A_k} S_k$ y $S_n - S_k$ son independientes

$$\implies \mathbb{E}[\mathbb{1}_{A_k(t)} S_k \cdot (S_n - S_k)] = \mathbb{E}[\mathbb{1}_{A_k(t)} S_k] \cdot \mathbb{E}[S_n - S_k] = 0. \quad \blacksquare$$